Chapter 40 Review

Campbell Biology, AP Edition, 12e

Chapter 40 Review

Summary of Key Concepts

Why are most animal homeostatic mechanisms controlled by negative feedback? Is that a characteristic of all organisms? **(BIG IDEA 2)**

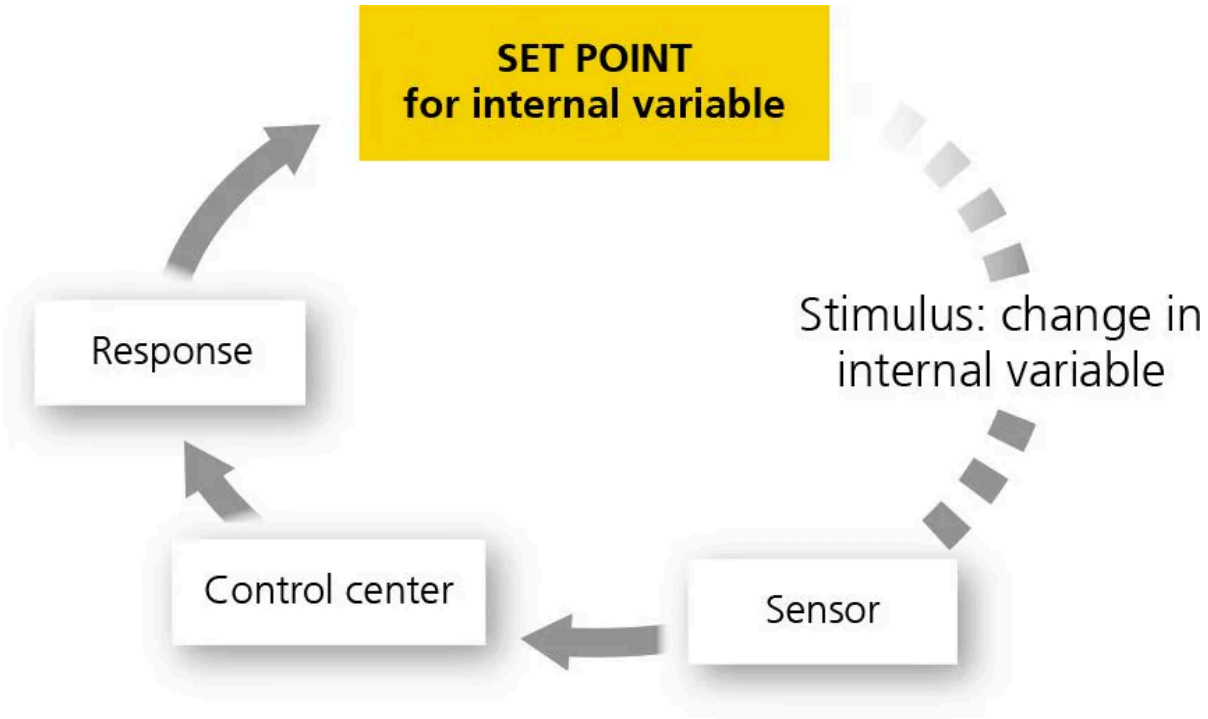
Concept 40.1: Animal form and function are correlated at all levels of organization

- Physical laws constrain the evolution of an animal's size and shape. These constraints contribute to convergent evolution in animal body forms.
- Each animal cell must have access to an aqueous environment. Simple two-layered sacs and flat shapes maximize exposure to the surrounding medium. More complex body plans have highly folded internal surfaces specialized for exchanging materials.
- Animal bodies are based on a hierarchy of cells, ****__tissues__, ****__organs__, and ****__organ systems__. **Epithelial tissue** forms active interfaces on external and internal surfaces; **connective tissue** binds and supports other tissues; **muscle tissue** contracts, moving body parts; and **nervous tissue** transmits nerve impulses throughout the body.
- The **endocrine** and **nervous systems** are the two means of communication between different locations in the body. The endocrine system broadcasts signaling molecules called ****__hormones__ everywhere via the bloodstream, but only certain cells are responsive to each hormone. The nervous system uses dedicated cellular circuits involving electrical and chemical signals to send information to specific locations.

For a large animal, what challenges would a spherical shape pose for carrying out exchange with the environment?

Concept 40.2: Feedback control maintains the internal environment in many animals

- An animal is a **regulator** if it controls an internal variable and a **conformer** if it allows an internal variable to vary with external changes. **Homeostasis** is the maintenance of a steady state despite internal and external changes.
- Homeostatic mechanisms are usually based on **negative feedback**, in which the **response** reduces the **stimulus**. In contrast, **positive feedback** involves amplification of a stimulus by the response and often brings about a change in state, such as the transition from pregnancy to childbirth.



A cyclical illustration shows how the process of negative feedback works. An internal variable has a set point. Due to a stimulus, a change in this set point is detected by a sensor and it sends the information to the control cortex. An appropriate response is made to return the variable to its set point. Arrows are present between categories: normal range for an internal variable to stimulus (change in internal variable) to sensor to control center to response back to normal range.

- Regulated change in the internal environment is essential to normal function. **********Circadian rhythms** are daily fluctuations in metabolism and behavior tuned to the cycles of light and dark in the environment. Other environmental changes may trigger **********acclimatization**, a temporary shift in the steady state.

Is it accurate to define homeostasis as a constant internal environment? Explain.

Concept 40.3: Homeostatic processes for thermoregulation involve form, function, and behavior

- An animal maintains its internal temperature within a tolerable range by **thermoregulation**. **********Endothermic** animals are warmed mostly by heat generated by metabolism. **Ectothermic** animals get most of their heat from external sources. Endothermy requires a greater expenditure of energy. Body temperature may vary with environmental temperature, as in *poikilotherms*, or be relatively constant, as in *homeotherms*.
- In thermoregulation, physiological and behavioral adjustments balance heat gain and loss, which occur through radiation, evaporation, convection, and conduction. Insulation and

****countercurrent exchange** reduce heat loss, whereas panting, sweating, and bathing increase evaporation, cooling the body. Many ectotherms and endotherms adjust their rate of heat exchange with their surroundings by vasodilation or vasoconstriction and by behavioral responses.

- Many mammals and birds adjust their amount of body insulation in response to changes in environmental temperature. Ectotherms undergo a variety of changes at the cellular level to acclimatize to shifts in temperature.
- The **hypothalamus** acts as the thermostat in mammalian regulation of body temperature. Fever reflects a resetting of this thermostat to a higher set point in response to infection.

Given that humans thermoregulate, explain why your skin is cooler than your body core.

Concept 40.4: Energy requirements are related to animal size, activity, and environment

- Animals obtain chemical energy from food, storing it for short-term use in ATP. The total amount of energy used in a unit of time defines an animal's **metabolic rate**.
- Under similar conditions and for animals of the same size, the **basal metabolic rate** of endotherms is substantially higher than the **standard metabolic rate** of ectotherms. Minimum metabolic rate per gram is inversely related to body size among similar animals. Animals allocate energy for basal (or standard) metabolism, activity, homeostasis, growth, and reproduction.
- ****Torpor**, a state of decreased activity and metabolism, conserves energy during environmental extremes. Animals may enter torpor according to a circadian rhythm (daily torpor), in winter (**hibernation**), or in summer (estivation).

Why do small animals breathe more rapidly than large animals?